



image analysis

MPhil in Data Intensive Science

Thursday 7rd May 2026

Minor: Image Analysis

- Problem Sheet 1: Image transforms and classical image processing

Dr A. Biguri

*In this practical session you will learn to perform basic tasks with images,
and to use some of the standard image transforms.*

*Attempt **all** questions.*

*We will work through the questions during the practical session on the date
indicated above.*

*Solutions to these questions will be available on Moodle after such practical
session.*

Module 1: Image and noise manipulation

Question 1: Using (`skimage`), load and display a couple of images, e.g. `coins` and `cat` . Inspect the size, type, number of channels of each of them. e

Question 2: Lets try to segment the coins. As they are brighter than the background, thresholding may work.

Question 3: The thresholding is close, but not sufficient. Use morphological operation to get circles for each coin.

[<https://scikit-image.org/docs/stable/api/skimage.morphology.html>]

Question 4: Lets try to get rid of things that are not coins. For that, we need to be able to select individual things. How?

Question 5: Now that we have individual items, lets try to get rid of the ones that are not a coin. We can do that by computing each region properties.

[<https://scikit-image.org/docs/stable/api/skimage.measure.html#skimage.measure.regionprops>]

Question 6: Lets pick one of them. E.g. the biggest one.

Question 7: Use the label to extract the coin only from the original image. Lets also remove unnecessary blank space.

Question 8: Lets give the cat something to look at. Lets put the coin in front of its face, rotated 30 clockwise. In front of its eyes is approximately at [200,350]. Do **not** use inbuilt transform functions, but code it yourself.

Question 9: Can you do the same thing using inbuilt functions?
